Question		Marks
6(a)	The r.m.s value of an <u>a.c</u>. will give the same <u>heating effect on a resistor</u> as due to <u>a d.c. current</u> of the <u>same value.</u>	B1
6(b)(i)	Power = VI = 1200 W r.m.s current I = 1200 / 120 = 10 A Peak Current $I_o = 10\sqrt{(2)} = 14.1$ A	 C1 A1
6(b) (ii)	$I = I_o \sin (2\pi f\, t)$ $= 14.1 \sin (2\pi \times 0.50t)$ $= 14.1 \sin (100\, \pi t)$	 B1
6(c)	Secondary power = 600 + 1200 = 1800 W By conservation of energy, Input power = 1800 W $I_pV_p = 1800$ $I_p = 1800/(2.4\times10^3)$ $= 0.75$ A	 C1 A1
7(a)(i)	an electric current that periodically reverses its direction in a circuit (with a	B1